

NASA HW8

1 Server setup

在Server端VM設定

```
apt update

# 安裝 OpenLDAP
apt install -y slapd ldap-utils

# 配置 SLAPD
dpkg-reconfigure slapd
```

創建 LDIF 檔案

```
nano /root/base.ldif
```

```
dn: ou=people,dc=nasa,dc=csie,dc=ntu
objectClass: organizationalUnit
ou: people

dn: ou=group,dc=nasa,dc=csie,dc=ntu
objectClass: organizationalUnit
ou: group
```

```
# 匯入
ldapadd -x -D "cn=admin,dc=nasa,dc=csie,dc=ntu" -W -f /root/base.ldif
```

```
root@hw8-server:~# ldapsearch -x -H ldapi:/// -b dc=nasa,dc=csie,dc=ntu
# extended LDIF
#
# LDAPv3
# base <dc=nasa,dc=csie,dc=ntu> with scope subtree
# filter: (objectclass=*)
# requesting: ALL
#
# nasa.csie.ntu
dn: dc=nasa,dc=csie,dc=ntu
objectClass: top
objectClass: dcObject
objectClass: organization
o: nasa
dc: nasa

# people, nasa.csie.ntu
dn: ou=people,dc=nasa,dc=csie,dc=ntu
objectClass: organizationalUnit
ou: people

# group, nasa.csie.ntu
dn: ou=group,dc=nasa,dc=csie,dc=ntu
objectClass: organizationalUnit
ou: group

# search result
search: 2
result: 0 Success

# numResponses: 4
# numEntries: 3
```

2 Client Setup

```
# 安裝 OpenLDAP 客戶端工具
pacman -Sy --noconfirm openldap
```

```
vim /etc/openldap/ldap.conf

### 調整以下
BASE    dc=nasa,dc=csie,dc=ntu
URI     ldap://192.168.8.1
```

```
[root@hw8-client ~]# ldapsearch -x
# extended LDIF
#
# LDAPv3
# base <dc=nasa,dc=csie,dc=ntu> (default) with scope subtree
# filter: (objectclass=*)
# requesting: ALL
#
# nasa.csie.ntu
dn: dc=nasa,dc=csie,dc=ntu
objectClass: top
objectClass: dcObject
objectClass: organization
o: nasa
dc: nasa

# people, nasa.csie.ntu
dn: ou=people,dc=nasa,dc=csie,dc=ntu
objectClass: organizationalUnit
ou: people

# group, nasa.csie.ntu
dn: ou=group,dc=nasa,dc=csie,dc=ntu
objectClass: organizationalUnit
ou: group

# search result
search: 2
result: 0 Success
```

3 LDAPS

(a)

```
# 安裝工具
apt install -y gnutls-bin

# 創建憑證目錄
mkdir -p /etc/ldap/ssl
cd /etc/ldap/ssl

# 產生 CA 私鑰
certtool --generate-privkey --outfile ca-key.pem

# CA 憑證範本
cat > ca.info << 'EOF'
cn = NASA LDAP CA
ca
cert_signing_key
expiration_days = 3650
EOF

# 產生 CA 憑證
certtool --generate-self-signed \
  --load-privkey ca-key.pem \
  --template ca.info \
  --outfile ca-cert.pem

# 產生伺服器私鑰
certtool --generate-privkey --outfile ldap-key.pem

# 創建伺服器憑證範本
cat > ldap.info << 'EOF'
organization = NASA CSIE NTU
cn = 192.168.8.1
tls_www_server
encryption_key
signing_key
expiration_days = 3650
EOF

# 產生伺服器憑證
certtool --generate-certificate \
  --load-privkey ldap-key.pem \
  --load-ca-certificate ca-cert.pem \
  --load-ca-privkey ca-key.pem \
  --template ldap.info \
  --outfile ldap-cert.pem

# 檔案權限
chown openldap:openldap /etc/ldap/ssl/*.pem
chmod 600 /etc/ldap/ssl/*-key.pem
chmod 644 /etc/ldap/ssl/ca-cert.pem
chmod 644 /etc/ldap/ssl/ldap-cert.pem
```

```
# 配置 LDAP 使用 TLS
nano /root/tls.ldif
```

```
dn: cn=config
changetype: modify
add: olcTLSCACertificateFile
olcTLSCACertificateFile: /etc/ldap/ssl/ca-cert.pem
-
add: olcTLSCertificateFile
olcTLSCertificateFile: /etc/ldap/ssl/ldap-cert.pem
-
add: olcTLSCertificateKeyFile
olcTLSCertificateKeyFile: /etc/ldap/ssl/ldap-key.pem
```

```
# 匯入 TLS 配置
ldapmodify -Y EXTERNAL -H ldapi:/// -f /root/tls.ldif
```

```
nano /etc/ldap/ldap.conf

# 修改以下
TLS_CACERT /etc/ldap/ssl/ca-cert.pem
TLS_REQCERT allow
```

```
nano /etc/default/slapd

# 修改以下
SLAPD_SERVICES="ldap:/// ldapi:/// ldaps:///"
```

```
systemctl restart slapd
systemctl status slapd
```

(b)

```

root@hw8-server:/etc/ldap/ssl# ldapsearch -x -ZZ -b dc=nasa,dc=csie,dc=ntu
# extended LDIF
#
# LDAPv3
# base <dc=nasa,dc=csie,dc=ntu> with scope subtree
# filter: (objectclass=*)
# requesting: ALL
#
# nasa.csie.ntu
dn: dc=nasa,dc=csie,dc=ntu
objectClass: top
objectClass: dcObject
objectClass: organization
o: nasa
dc: nasa

# people, nasa.csie.ntu
dn: ou=people,dc=nasa,dc=csie,dc=ntu
objectClass: organizationalUnit
ou: people

# group, nasa.csie.ntu
dn: ou=group,dc=nasa,dc=csie,dc=ntu
objectClass: organizationalUnit
ou: group

# search result
search: 3
result: 0 Success

# numResponses: 4
# numEntries: 3

```

(c)

```

root@hw8-server:/etc/ldap/ssl# ldapsearch -x -H ldaps://192.168.8.1 -b dc=nasa,dc=csie,dc=ntu
# extended LDIF
#
# LDAPv3
# base <dc=nasa,dc=csie,dc=ntu> with scope subtree
# filter: (objectclass=*)
# requesting: ALL
#
# nasa.csie.ntu
dn: dc=nasa,dc=csie,dc=ntu
objectClass: top
objectClass: dcObject
objectClass: organization
o: nasa
dc: nasa

# people, nasa.csie.ntu
dn: ou=people,dc=nasa,dc=csie,dc=ntu
objectClass: organizationalUnit
ou: people

# group, nasa.csie.ntu
dn: ou=group,dc=nasa,dc=csie,dc=ntu
objectClass: organizationalUnit
ou: group

# search result
search: 2
result: 0 Success

# numResponses: 4
# numEntries: 3

```

(d)

```
nano /root/force-tls.ldif
```

```
dn: olcDatabase={1}mdb,cn=config  
changetype: modify  
add: olcSecurity  
olcSecurity: tls=1
```

匯入設定

```
ldapmodify -Y EXTERNAL -H ldapi:/// -f /root/force-tls.ldif
```

(e)

Server 端找出憑證

```
cat /etc/ldap/ssl/ca-cert.pem
```

在 client VM 上創建憑證

```
mkdir -p /etc/openldap/certs
```

```
cat > /etc/openldap/certs/ca-cert.pem << 'EOF'  
-----BEGIN CERTIFICATE-----  
MIID/jCCAmagAwIBAgIUeg ...  
-----END CERTIFICATE-----  
EOF
```

修改設定

```
vim /etc/openldap/ldap.conf  
# 加入這兩行  
TLS_CACERT /etc/openldap/certs/ca-cert.pem  
TLS_REQCERT allow
```

```
[root@hw8-client ~]# ldapsearch -x -H ldaps://192.168.8.1 -b dc=nasa,dc=csie,dc=ntu
# extended LDIF
#
# LDAPv3
# base <dc=nasa,dc=csie,dc=ntu> with scope subtree
# filter: (objectclass=*)
# requesting: ALL
#
# nasa.csie.ntu
dn: dc=nasa,dc=csie,dc=ntu
objectClass: top
objectClass: dcObject
objectClass: organization
o: nasa
dc: nasa

# people, nasa.csie.ntu
dn: ou=people,dc=nasa,dc=csie,dc=ntu
objectClass: organizationalUnit
ou: people

# group, nasa.csie.ntu
dn: ou=group,dc=nasa,dc=csie,dc=ntu
objectClass: organizationalUnit
ou: group

# search result
search: 2
result: 0 Success

# numResponses: 4
# numEntries: 3
```

(f)

```
[root@hw8-client ~]# ldapsearch -x -H ldap://192.168.8.1 -b dc=nasa,dc=csie,dc=ntu
# extended LDIF
#
# LDAPv3
# base <dc=nasa,dc=csie,dc=ntu> with scope subtree
# filter: (objectclass=*)
# requesting: ALL
#
# search result
search: 2
result: 13 Confidentiality required
text: TLS confidentiality required

# numResponses: 1
```

4

創建群組及使用者

```
cat > /root/groups.ldif << 'EOF'
dn: cn=ta,ou=group,dc=nasa,dc=csie,dc=ntu
objectClass: posixGroup
cn: ta
gidNumber: 10001

dn: cn=student,ou=group,dc=nasa,dc=csie,dc=ntu
objectClass: posixGroup
cn: student
gidNumber: 10002
EOF
```

匯入群組

```
ldapadd -x -ZZ -D "cn=admin,dc=nasa,dc=csie,dc=ntu" -W -f /root/groups.ldif
```

```
# ta
slappasswd
{SSHA}6M79iBrP5tuin0LM956tB2+5NI1KI8wm
# student
slappasswd
{SSHA}KZagpb0kq93bl5BYRt59QKBDaZSWyZrV
```

```
cat > /root/users.ldif << 'EOF'
dn: uid=ta001,ou=people,dc=nasa,dc=csie,dc=ntu
objectClass: inetOrgPerson
objectClass: posixAccount
objectClass: shadowAccount
uid: ta001
sn: TA
givenName: Teaching
cn: Teaching Assistant
displayName: TA 001
uidNumber: 20001
gidNumber: 10001
userPassword: {SSHA}6M79iBrP5tuin0LM956tB2+5NI1KI8wm
loginShell: /bin/bash
homeDirectory: /home/ta001
mail: ta001@nasa.csie.ntu

dn: uid=student001,ou=people,dc=nasa,dc=csie,dc=ntu
objectClass: inetOrgPerson
objectClass: posixAccount
objectClass: shadowAccount
uid: student001
sn: Student
givenName: Normal
cn: Normal Student
displayName: Student 001
uidNumber: 20002
gidNumber: 10002
userPassword: {SSHA}KZagpb0kq93bl5BYRt59QKBDaZSWyZrV
loginShell: /bin/bash
homeDirectory: /home/student001
mail: student001@nasa.csie.ntu
EOF
```

匯入使用者：

```
ldapadd -x -ZZ -D "cn=admin,dc=nasa,dc=csie,dc=ntu" -W -f /root/users.ldif
```

sudo 規則

在 Server VM 上：

```
cat > /root/sudo-schema.ldif << 'EOF'
dn: cn=sudo,cn=schema,cn=config
objectClass: olcSchemaConfig
cn: sudo
olcAttributeTypes: ( 1.3.6.1.4.1.15953.9.1.1 NAME 'sudoUser' DESC 'User(s) who may run sudo'
EQUALITY caseExactMatch SUBSTR caseExactSubstringsMatch SYNTAX 1.3.6.1.4.1.1466.115.121.1.15
)
olcAttributeTypes: ( 1.3.6.1.4.1.15953.9.1.2 NAME 'sudoHost' DESC 'Host(s) who may run sudo'
EQUALITY caseExactMatch SUBSTR caseExactSubstringsMatch SYNTAX 1.3.6.1.4.1.1466.115.121.1.15
)
olcAttributeTypes: ( 1.3.6.1.4.1.15953.9.1.3 NAME 'sudoCommand' DESC 'Command(s) to be execut
ed by sudo' EQUALITY caseExactMatch SYNTAX 1.3.6.1.4.1.1466.115.121.1.15 )
olcAttributeTypes: ( 1.3.6.1.4.1.15953.9.1.4 NAME 'sudoRunAs' DESC 'User(s) impersonated by s
udo' EQUALITY caseExactMatch SYNTAX 1.3.6.1.4.1.1466.115.121.1.15 )
olcAttributeTypes: ( 1.3.6.1.4.1.15953.9.1.5 NAME 'sudoOption' DESC 'Options(s) followed by s
udo' EQUALITY caseExactMatch SYNTAX 1.3.6.1.4.1.1466.115.121.1.15 )
olcAttributeTypes: ( 1.3.6.1.4.1.15953.9.1.6 NAME 'sudoRunAsUser' DESC 'User(s) impersonated
by sudo' EQUALITY caseExactMatch SYNTAX 1.3.6.1.4.1.1466.115.121.1.15 )
olcAttributeTypes: ( 1.3.6.1.4.1.15953.9.1.7 NAME 'sudoRunAsGroup' DESC 'Group(s) impersonate
d by sudo' EQUALITY caseExactMatch SYNTAX 1.3.6.1.4.1.1466.115.121.1.15 )
olcAttributeTypes: ( 1.3.6.1.4.1.15953.9.1.8 NAME 'sudoNotBefore' DESC 'Start of time interva
l for sudoers entry' EQUALITY generalizedTimeMatch ORDERING generalizedTimeOrderingMatch SYNT
AX 1.3.6.1.4.1.1466.115.121.1.24 )
olcAttributeTypes: ( 1.3.6.1.4.1.15953.9.1.9 NAME 'sudoNotAfter' DESC 'End of time interval f
or sudoers entry' EQUALITY generalizedTimeMatch ORDERING generalizedTimeOrderingMatch SYNTAX
1.3.6.1.4.1.1466.115.121.1.24 )
olcAttributeTypes: ( 1.3.6.1.4.1.15953.9.1.10 NAME 'sudoOrder' DESC 'Ordering of sudoers entr
ies' EQUALITY integerMatch ORDERING integerOrderingMatch SYNTAX 1.3.6.1.4.1.1466.115.121.1.27
)
olcObjectClasses: ( 1.3.6.1.4.1.15953.9.2.1 NAME 'sudoRole' DESC 'Sudoers Role' SUP top STRUC
TURAL MUST cn MAY ( sudoUser $ sudoHost $ sudoCommand $ sudoRunAs $ sudoRunAsUser $ sudoRunAs
Group $ sudoOption $ sudoOrder $ sudoNotBefore $ sudoNotAfter $ description ) )
EOF
```

匯入

```
ldapadd -Y EXTERNAL -H ldapi:/// -f /root/sudo-schema.ldif
```

TA 加入 sudo

```
cat > /root/sudo-rules.ldif << 'EOF'
dn: ou=sudoers,dc=nasa,dc=csie,dc=ntu
objectClass: organizationalUnit
ou: sudoers

dn: cn=ta-sudo,ou=sudoers,dc=nasa,dc=csie,dc=ntu
objectClass: sudoRole
cn: ta-sudo
sudoUser: %ta
sudoHost: ALL
sudoCommand: ALL
sudoRunAsUser: ALL
EOF
```

匯入

```
ldapadd -x -D "cn=admin,dc=nasa,dc=csie,dc=ntu" -W -f /root/sudo-rules.ldif
```

在 *Client* 安裝和設定 *SSSD*

安裝套件

```
pacman -Sy --noconfirm sssd sudo
```

建立 `/etc/sss/sss.conf`

```
cat > /etc/sss/sss.conf << 'EOF'
[sss]
services = nss, pam, sudo
config_file_version = 2
domains = nasa.csie.ntu

[domain/nasa.csie.ntu]
id_provider = ldap
auth_provider = ldap
sudo_provider = ldap
ldap_uri = ldaps://192.168.8.1
ldap_search_base = dc=nasa,dc=csie,dc=ntu
ldap_sudo_search_base = ou=sudoers,dc=nasa,dc=csie,dc=ntu
ldap_tls_cacert = /etc/openldap/certs/ca-cert.pem
ldap_tls_reqcert = allow
enumerate = True
cache_credentials = True
EOF

chmod 600 /etc/sss/sss.conf
```

修改 `/etc/nsswitch.conf`

```
passwd: files sss
group: files sss
shadow: files sss
sudoers: files sss
hosts: files dns
```

設定 PAM (system-auth)

```
cat > /etc/pam.d/system-auth << 'EOF'
#%PAM-1.0

auth      sufficient      pam_sss.so      forward_pass
auth      required        pam_unix.so     try_first_pass nullok
auth      optional        pam_permit.so

account   sufficient      pam_sss.so
account   required        pam_unix.so
account   optional        pam_permit.so

password  sufficient      pam_sss.so      use_authtok
password  required        pam_unix.so     try_first_pass nullok sha512 shado
w
password  optional        pam_permit.so

session   optional        pam_keyinit.so   revoke
session   required        pam_limits.so
session   required        pam_unix.so
session   optional        pam_sss.so
session   required        pam_mkhomedir.so skel=/etc/skel umask=0077
EOF
```

同時修改 sshd 專用的 PAM

```
cat > /etc/pam.d/sshd << 'EOF'
#%PAM-1.0

auth      include      system-remote-login
account   include      system-remote-login
password  include      system-remote-login
session   include      system-remote-login
```

system-remote-login

```
cat > /etc/pam.d/system-remote-login << 'EOF'
#%PAM-1.0

auth      sufficient pam_sss.so      forward_pass
auth      include    system-login

account   sufficient pam_sss.so
account   include    system-login

password  include    system-login

session   optional   pam_loginuid.so
session   include    system-login
session   optional   pam_mkhomedir.so skel=/etc/skel umask=0077
EOF
```

system-login

```
cat > /etc/pam.d/system-login << 'EOF'
#%PAM-1.0

auth      required    pam_tally2.so      onerr=succeed file=/var/log/tallylog
auth      required    pam_shells.so
auth      requisite   pam_nologin.so
auth      sufficient   pam_sss.so
auth      required    pam_unix.so      try_first_pass nullok

account   required    pam_access.so
account   required    pam_nologin.so
account   sufficient   pam_sss.so
account   required    pam_unix.so
account   required    pam_permit.so

password  sufficient   pam_sss.so      use_authok
password  required    pam_unix.so      try_first_pass nullok sha512 shadow

session   required    pam_limits.so
session   required    pam_unix.so
session   optional    pam_sss.so
session   optional    pam_mkhomedir.so skel=/etc/skel umask=0077
EOF
```

重启

```
systemctl restart sshd
```

```

b13705020@nasaws3:~$ ssh -p 13706 ta001@localhost
ta001@localhost's password:
Creating directory '/home/ta001'.
[ta001@hw8-client ~]$ whoami
id
ta001
uid=20001(ta001) gid=10001(ta) groups=10001(ta)
[ta001@hw8-client ~]$ sudo whoami

We trust you have received the usual lecture from the local System
Administrator. It usually boils down to these three things:

    #1) Respect the privacy of others.
    #2) Think before you type.
    #3) With great power comes great responsibility.

For security reasons, the password you type will not be visible.

[sudo] password for ta001:
root
[ta001@hw8-client ~]$ whoami
ta001
[ta001@hw8-client ~]$ sudo echo "Hello World"
Hello World

```

```

b13705020@nasaws3:~$ ssh -p 13706 student001@localhost
student001@localhost's password:
Creating directory '/home/student001'.
[student001@hw8-client ~]$ whoami
id
student001
uid=20002(student001) gid=10002(student) groups=10002(student)
[student001@hw8-client ~]$ sudo whoami

We trust you have received the usual lecture from the local System
Administrator. It usually boils down to these three things:

    #1) Respect the privacy of others.
    #2) Think before you type.
    #3) With great power comes great responsibility.

For security reasons, the password you type will not be visible.

[sudo] password for student001:
student001 is not allowed to run sudo on hw8-client.
[student001@hw8-client ~]$ sudo echo "Hello World"
[sudo] password for student001:
student001 is not allowed to run sudo on hw8-client.

```

5

新的 ACL 規則

```
nano /root/acl.ldif
```

```

dn: olcDatabase={1}mdb,cn=config
changetype: modify
delete: olcAccess
-
add: olcAccess
olcAccess: {0}to attrs=userPassword
    by self write
    by anonymous auth
    by * none
-
add: olcAccess
olcAccess: {1}to attrs=cn,uid,uidNumber,gidNumber,homeDirectory
    by self read
    by * read
-
add: olcAccess
olcAccess: {2}to *
    by self write
    by users read
    by anonymous read

```

規則 0 (userPassword):

- `by self write` - 使用者可以修改自己的密碼
- `by anonymous auth` - 匿名用戶可以用來認證
- `by * none` - 其他人完全不能讀取

規則 1 (不可修改的屬性):

- `by self read` - 使用者只能讀取自己的這些屬性
- `by * read` - 其他人可以讀取

規則 2 (其他所有屬性):

- `by self write` - 使用者可以修改自己的
- `by users read` - 已認證使用者可以讀取
- `by anonymous read` - 匿名也可以讀取

匯入

```
ldapmodify -Y EXTERNAL -H ldapi:/// -f /root/acl.ldif
```

(a)

以 ta001 身份嘗試修改 student001，失敗

```

root@hw8-server:~# cat > /root/test-modify-other.ldif << 'EOF'
dn: uid=student001,ou=people,dc=nasa,dc=csie,dc=ntu
changetype: modify
replace: loginShell
loginShell: /bin/zsh
EOF
root@hw8-server:~# ldapmodify -x -D "uid=ta001,ou=people,dc=nasa,dc=csie,dc=ntu" -W -H ldaps://192.168.8.1 -f /root/test-modify-other.ldif
Enter LDAP Password:
modifying entry "uid=student001,ou=people,dc=nasa,dc=csie,dc=ntu"
ldap_modify: Insufficient access (50)

```

(b)

修改允許的屬性 (loginShell) - 成功

```

root@hw8-server:~# cat > /root/test-modify-self-allowed.ldif << 'EOF'
dn: uid=ta001,ou=people,dc=nasa,dc=csie,dc=ntu
changetype: modify
replace: loginShell
loginShell: /bin/zsh
EOF
root@hw8-server:~# ldapmodify -x -D "uid=ta001,ou=people,dc=nasa,dc=csie,dc=ntu" -W -H ldaps://192.168.8.1 -f /root/test-modify-self-allowed.ldif
Enter LDAP Password:
modifying entry "uid=ta001,ou=people,dc=nasa,dc=csie,dc=ntu"

```

修改受限屬性 (uidNumber) - 失敗

```

root@hw8-server:~# cat > /root/test-modify-self-denied.ldif << 'EOF'
dn: uid=ta001,ou=people,dc=nasa,dc=csie,dc=ntu
changetype: modify
replace: uidNumber
uidNumber: 99999
EOF
root@hw8-server:~# ldapmodify -x -D "uid=ta001,ou=people,dc=nasa,dc=csie,dc=ntu" -W -H ldaps://192.168.8.1 -f /root/test-modify-self-denied.ldif
Enter LDAP Password:
modifying entry "uid=ta001,ou=people,dc=nasa,dc=csie,dc=ntu"
ldap_modify: Insufficient access (50)

```

(c)

以 ta001 身份讀取 student001 的資料：

```

root@hw8-server:~# ldapsearch -x -D "uid=ta001,ou=people,dc=nasa,dc=csie,dc=ntu" -W -H ldaps://192.168.8.1 -b "uid=student001,ou=people,dc=nasa,dc=csie,dc=ntu"
Enter LDAP Password:
# extended LDIF
#
# LDAPv3
# base <uid=student001,ou=people,dc=nasa,dc=csie,dc=ntu> with scope subtree
# filter: (objectclass=*)
# requesting: ALL
#
# student001, people, nasa.csie.ntu
dn: uid=student001,ou=people,dc=nasa,dc=csie,dc=ntu
objectClass: inetOrgPerson
objectClass: posixAccount
objectClass: shadowAccount
uid: student001
sn: Student
givenName: Normal
cn: Normal Student
displayName: Student 001
uidNumber: 20002
gidNumber: 10002
loginShell: /bin/bash
homeDirectory: /home/student001
mail: student001@nasa.csie.ntu

# search result
search: 2
result: 0 Success

# numResponses: 2
# numEntries: 1
root@hw8-server:~#

```

匿名讀取

```
root@hw8-server:~# ldapsearch -x -H ldaps://192.168.8.1 -b "uid=student001,ou=people,dc=nasa,dc=csie,dc=ntu"
# extended LDIF
#
# LDAPv3
# base <uid=student001,ou=people,dc=nasa,dc=csie,dc=ntu> with scope subtree
# filter: (objectclass=*)
# requesting: ALL
#
# student001, people, nasa.csie.ntu
dn: uid=student001,ou=people,dc=nasa,dc=csie,dc=ntu
objectClass: inetOrgPerson
objectClass: posixAccount
objectClass: shadowAccount
uid: student001
sn: Student
givenName: Normal
cn: Normal Student
displayName: Student 001
uidNumber: 20002
gidNumber: 10002
loginShell: /bin/bash
homeDirectory: /home/student001
mail: student001@nasa.csie.ntu
# search result
search: 2
result: 0 Success
# numResponses: 2
# numEntries: 1
```

6

先備份

```
# 停止服務
systemctl stop slapd

# 備份資料庫
cp -r /var/lib/ldap /var/lib/ldap.backup

# 備份設定
cp -r /etc/ldap /etc/ldap.backup

# 重啟服務
systemctl start slapd
```

創建 Schema

```

cat > /root/ntu-student-schema.ldif << 'EOF'
dn: cn=ntustudent,cn=schema,cn=config
objectClass: olcSchemaConfig
cn: ntustudent
olcAttributeTypes: ( 1.3.6.1.4.1.99999.1.1.1
    NAME 'studentEntryMethod'
    DESC 'Student entry method to NTU'
    EQUALITY caseIgnoreMatch
    SUBSTR caseIgnoreSubstringsMatch
    SYNTAX 1.3.6.1.4.1.1466.115.121.1.15
    SINGLE-VALUE )
olcAttributeTypes: ( 1.3.6.1.4.1.99999.1.1.2
    NAME 'studentAdvisor'
    DESC 'Student advisor name'
    EQUALITY caseIgnoreMatch
    SUBSTR caseIgnoreSubstringsMatch
    SYNTAX 1.3.6.1.4.1.1466.115.121.1.15
    SINGLE-VALUE )
olcAttributeTypes: ( 1.3.6.1.4.1.99999.1.1.3
    NAME 'studentClub'
    DESC 'Student club membership'
    EQUALITY caseIgnoreMatch
    SUBSTR caseIgnoreSubstringsMatch
    SYNTAX 1.3.6.1.4.1.1466.115.121.1.15 )
olcObjectClasses: ( 1.3.6.1.4.1.99999.1.2.1
    NAME 'ntuStudent'
    DESC 'NTU Student'
    SUP top
    AUXILIARY
    MUST ( studentEntryMethod $ studentAdvisor )
    MAY ( studentClub ) )
EOF

```

定義：

1. `studentEntryMethod` - 入學管道（必要）
2. `studentAdvisor` - 導師（必要）
3. `studentClub` - 社團（可選，可多個）

objectClass：

- `ntuStudent` - *AUXILIARY* 類型（輔助類型，可以加到現有使用者上）
- *MUST*：必須有 `studentEntryMethod` 和 `studentAdvisor`
- *MAY*：可以有多個 `studentClub`

匯入

```

ldapadd -Y EXTERNAL -H ldapi:/// -f /root/ntu-student-schema.ldif

```

創建使用者

雜湊密碼

```

STUDENT_PASS=$(slappasswd -s student002pass)
echo "Student002 password hash: $STUDENT_PASS"

```

創建使用者資訊

```
cat > /root/ntu-student-user.ldif << EOF
dn: uid=student002,ou=people,dc=nasa,dc=csie,dc=ntu
objectClass: inetOrgPerson
objectClass: posixAccount
objectClass: shadowAccount
objectClass: ntuStudent
uid: student002
sn: Chen
givenName: Tim
cn: Chen Tim
displayName: Student 002
uidNumber: 20003
gidNumber: 10002
userPassword: $STUDENT_PASS
loginShell: /bin/bash
homeDirectory: /home/student002
mail: student002@nasa.csie.ntu
studentEntryMethod: self_apply
studentAdvisor: hsinmu
studentClub: aiclub
studentClub: imclub
EOF
```

匯入

```
ldapadd -x -ZZ -D "cn=admin,dc=nasa,dc=csie,dc=ntu" -W -f /root/ntu-student-user.ldif
```

```
root@hw8-server:~# ldapsearch -x -H ldaps://192.168.8.1 -b "uid=student002,ou=people,dc=nasa,dc=csie,dc=ntu"
''
# extended LDIF
#
# LDAPv3
# base <uid=student002,ou=people,dc=nasa,dc=csie,dc=ntu> with scope subtree
# filter: (objectclass=*)
# requesting: ALL
#
# student002, people, nasa.csie.ntu
dn: uid=student002,ou=people,dc=nasa,dc=csie,dc=ntu
objectClass: inetOrgPerson
objectClass: posixAccount
objectClass: shadowAccount
objectClass: ntuStudent
uid: student002
sn: Chen
givenName:: VGltIA==
cn: Chen Tim
displayName: Student 002
uidNumber: 20003
gidNumber: 10002
loginShell: /bin/bash
homeDirectory: /home/student002
mail: student002@nasa.csie.ntu
studentEntryMethod: self_apply
studentAdvisor: hsinmu
studentClub: aiclub
studentClub: imclub

# search result
search: 2
result: 0 Success

# numResponses: 2
# numEntries: 1
```